

PAPER_531 — Big Bang Hypergraph Origin & VDS Partition Function Emergence

Unified Quantum Field Framework (UQFF)

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§1 — Overview

This paper establishes the UQFF description of the **Big Bang as the first Wolfram hypergraph rewrite step** applied to the seed graph G_0 . The Superconductor Mediator $\text{SCm}(t)$ provides a continuous observer-time measure of cosmic expansion, and the **Vacuum Density Series (VDS)** partition function $Z = \text{Li}_{26}(\text{SSq})$ emerges analytically as the generating function of distinguishable causal bonds at the 26-dimensional projection limit.

§2 — SCm Expansion Equation

$$\text{SCm}(t) = \lambda_{\text{ua}} \cdot U_{\text{UA}} \cdot \left(1 - \frac{1}{t}\right)$$

At the first rewrite step $t = 1$: $\text{SCm}(1) = 0$ (no vacuum mediator yet). As $t \rightarrow \infty$: $\text{SCm} \rightarrow \lambda_{\text{ua}} \cdot U_{\text{UA}}$ (vacuum ground state).

The **VDS partition function** is:

$$Z = \text{Li}_{26}(\text{SSq}) = \sum_{k=1}^{26} \frac{(\text{SSq})^k}{k^{26}} \approx 0.5699$$

This is a 26-term Lerch transcendent evaluated at $[\text{SSq}] = 0.57$, representing the total number of distinguishable vacuum microstates at the 26D projection boundary of the Wolfram hypergraph.

§3 — Causal Graph Growth

Each Wolfram rewrite step adds one causal node:

$$|V(G_n)| = n + 1$$

The rewrite count at the current cosmological epoch ($t_0 = 4.35 \times 10^{17}$ s):

$$n_0 = \frac{t_0}{\tau_{\text{Planck}}} = \frac{4.35 \times 10^{17}}{5.39 \times 10^{-44}} \approx 8.07 \times 10^{60}$$

The VDS series builds combinatorially at each step; by step $n \gg 1$ it has converged fully to $Z \approx 0.5699$.

§4 — CMB ISW Prediction

The angular power ratio in the CMB temperature spectrum:

$$\frac{C_{26}}{C_{22}} = \frac{[\text{SSq}]^{26} / 26^{26}}{[\text{SSq}]^{22} / 22^{26}} \approx 1.8 \times 10^{-3}$$

This predicts a VDS-driven excess at multipole $\ell = 26$ relative to $\ell = 22$ in the Planck 2018 TT spectrum, consistent with the observed ISW angular power at $\ell \sim 26$.

Observable	UQFF Prediction	Planck 2018
$\ell = 26$ excess	$C_{26}/C_{22} \approx 1.8 \times 10^{-3}$	$\sim 2 \times 10^{-3}$ (ISW plateau)
$\text{SCm}(t_0)$	$\approx 9.997 \times 10^{-5}$	$U_{\text{UA}} \sim 10^{-4}$ (canonical)
VDS convergence Z	0.5699 (26 terms)	— (theoretical prediction)

§5 — Entropy Ratchet

The Wolfram rewrite is **irreversible**: each application increases the causal graph by exactly one node. The entropy:

$$S(n) = n \quad \text{quod} \quad \text{monotone; measures causal graph complexity}$$

This establishes the **cosmological arrow of time** as a direct consequence of the Big Bang hypergraph rewrite irreversibility — no external time-asymmetry assumption is required.

§6 — Connection to UQFF Equilibrium

At $\text{SCm} = \lambda_{\text{ua}} \cdot U_{\text{UA}}$ (cosmic equilibrium):

$$F_U = U_g + U_m + U_b = 0$$

The field reaches full encompassment. Z being nonzero and finite (> 0) ensures that the VDS partition function never vanishes, preventing the Boltzmann factor $e^{-E/F_{\text{max}}}/Z$ from diverging — a necessary condition for the Yang-Mills mass gap ($\Delta > 0$, PAPER_540).

§7 — Available Equations

Equation	Description		
$\text{SCm}(t) = \lambda_{\text{ua}} \cdot U_{\text{UA}} \cdot (1 - 1/t)$	Expansion mediator		
$Z = \sum_{k=1}^{26} [\text{SSq}]^k / k^{26}$	VDS partition function		
n_0	$V(G_n)$	$= n+1$	Causal graph node count
t_0/τ_{Planck}	Rewrite count at present epoch		
$C_{26}/C_{22} = ([\text{SSq}]^{26}/26^{26})/([\text{SSq}]^{22}/22^{26})$	CMB ISW ratio		
$F_U = 0$ at $\text{SCm} = \lambda_{\text{ua}} U_{\text{UA}}$	UQFF equilibrium condition		

§8 — CP4 Calculator Output

```
calc = BigBangHypergraphOriginCalculator()
result = calc.compute()
```

```
# result[&#39;SCm_now&#39;]      - current SCm value (~9.997e-5)
# result[&#39;VDS_Z&#39;]        - partition function Z ( $\approx 0.5699$ )
# result[&#39;CMB_C26_C22&#39;]  - CMB ISW power ratio
# result[&#39;SCm_vacuum_lim&#39;] - vacuum state asymptote
```

§9 — References

- PAPER_429: Three New UQFF Number Systems (VDS · DVP · BH)
- PAPER_528: UQFF_comp Spectral Compression Eigenvalue Stability
- PAPER_540: Yang-Mills DPM Gauge Quantization (Z denominates gap)
- grok_share_fd81483544d.txt: BigBangHypergraphTheory source document
- Wolfram (2002): A New Kind of Science — hypergraph rewrite foundations
- Planck Collaboration (2018): TT, TE, EE power spectra